

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

## ASSESSMENT TOOL 1 – CODING GOLF COMPETITION

|                  |                                                                                                     |               |                                             |
|------------------|-----------------------------------------------------------------------------------------------------|---------------|---------------------------------------------|
| Course Code:     | CSA0309                                                                                             | Course Title: | Data Structures                             |
| CO Assessed:     | CO4 – Articulation (BL3)                                                                            | Assessment:   | Assessment Tool 1 – Coding Golf Competition |
| Weightage:       | 35% of CIA                                                                                          | Total Marks:  | 100                                         |
| CO4 Statement:   | Apply hashing strategies for efficient data storage and sorting algorithms for arrangement of data. |               |                                             |
| Student Name:    | G. Nithya Sree                                                                                      | Reg. No.:     | 192424103                                   |
| Section / Batch: | 3 <sup>RD</sup> Year AIDS                                                                           | Date:         | 14/08/2024                                  |

### Code of Conduct

*I G. Nithya Sree (192424103) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: G. Nithya Sree

### Instructions

- Log in to the assigned coding platform - Code chef (<https://www.codechef.com/skill-test/c-language-dsa>). Total: 100 Marks.
- You can only attempt this assessment once. Also, you will not be able to pause the assessment after starting.
- You are not allowed to switch tabs during the assessment, otherwise you will be blocked from the assessment.
- You will get a detailed report on your performance at the end of the assessment.

| Section / Topic | Focus     | Q Nos     | Marks |
|-----------------|-----------|-----------|-------|
| Data structures | Code Chef | 28        | 100   |
| GRAND TOTAL:    |           | 100 Marks |       |

## Rubrics for Calculation

| Criteria                                        | Weightage | Level 4 – Exemplary                                                                         | Level 3 – Proficient                                       | Level 2 – Developing                                        | Level 1 – Beginning                                           |
|-------------------------------------------------|-----------|---------------------------------------------------------------------------------------------|------------------------------------------------------------|-------------------------------------------------------------|---------------------------------------------------------------|
| <b>Problem Understanding</b>                    | 20        | Accurately interprets all problem requirements and constraints before coding.               | Understands most requirements with minor omissions.        | Partially understands the problem; misses some constraints. | Misinterprets the problem or ignores key requirements.        |
| <b>Correctness of Solution</b>                  | 30        | Program compiles successfully and passes all hidden and visible test cases.                 | Program passes most test cases with minor errors.          | Program passes some test cases but has logical issues.      | Program fails to compile or produces incorrect results.       |
| <b>Data Structure &amp; Algorithm Selection</b> | 20        | Selects the most appropriate data structure and algorithm with an optimized solution.       | Chooses an appropriate solution with minor inefficiencies. | Uses a workable but less efficient approach.                | Uses an inappropriate data structure or algorithm.            |
| <b>Code Quality &amp; Documentation</b>         | 20        | Well-structured, modular, readable, and properly indented code with meaningful identifiers. | Readable code with minor formatting issues.                | Basic coding style with inconsistent formatting.            | Poorly organized code that is difficult to understand.        |
| <b>Time &amp; Space Optimization</b>            | 10        | Demonstrates excellent optimization and correctly analyses time and space complexity.       | Good optimization with minor inefficiencies.               | Limited optimization and incomplete complexity analysis.    | No optimization and incorrect or missing complexity analysis. |

| Faculty In-charge | Course Coordinator | Program Director |
|-------------------|--------------------|------------------|
| Signature: _____  | Signature: _____   | Signature: _____ |
| Date: _____       | Date: _____        | Date: _____      |



CODECHEF Skill Tests

### Data structures and Algorithms in C test

Nithya Sree Golamary scored

**1400** /3000 points

● **14** Correct Answers
 ● **02** Wrong Answers
 ● **14** Unattempted



**46%**

SKILL SCORE